



Cloud Security with AWS IAM



Amir Hamzah

Policy editor

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Action": "ec2:*",
7       "Resource": "*",
8       "Condition": {
9         "StringEquals": {
10          "ec2:ResourceTag/Env": "development"
11        }
12      }
13    },
14    {
15      "Effect": "Allow",
16      "Action": "ec2:Describe*",
17      "Resource": "*"
18    },
19    {
20      "Effect": "Deny",
21      "Action": [
22        "ec2:DeleteTags",
23        "ec2:CreateTags"
24      ],
25      "Resource": "*"
26    }
27  ]
28 }
29
```



Introducing Today's Project!

In this project, I will demonstrate how to use AWS IAM to control access and permission settings in our AWS account. I'm doing this project to learn about cloud security from the absolute foundations.

Tools and concepts

Services I used were Amazon EC2 and AWS IAM. Key concepts I learnt include IAM users, policies, user groups and account aliases. I also learn how to use the Policy Simulator and how JSON policies work. How to launch EC2, tag an instance and login user.

Project reflection

This project took me approximately 1.5 hours today including project demo time. The most challenging part was understanding the IAM policies. It was most rewarding for me to see permission denied when intern tried to delete production instance.



Tags

Tags are organisational tools that lets us label our resources.They are helpful for grouping resources, cost allocation and applying policies for all resources with the same tag.

The tags I've used on my EC2 instances is called Env, which stands for environment. The value we've assigned for our instances are production and development!

▼ Name and tags [Info](#)

Key [Info](#)

Q Name X

Value [Info](#)

Q nextwork-dev-amirhar X

Resource types [Info](#)

Select resource types ▼

Instances X

Remove

Key [Info](#)

Q Env X

Value [Info](#)

Q development X

Resource types [Info](#)

Select resource types ▼

Instances X

Remove

Add new tag

You can add up to 48 more tags.



IAM Policies

IAM Policies are like rules that determine who can do what in an AWS account. I'm using policies today to control who has access to our production/environment instance.

The policy I set up

For this project, I've set up a policy using JSON.

I've created a policy that allows the policy holder to have permission to do anything they want to any instance tagged with "development". They can also see information for any instance, but they're denied access to deleting/creating tags for any inst

When creating a JSON policy, you have to define its Effect, Action and Resource.

The Effect, Action, and Resource attributes of a JSON policy means whether or not the policy is allowing/denying action; what the policy holder can or cannot do; and the specific AWS resources that the policy relates to.



My JSON Policy

```
Policy editor
1  {
2    "Version": "2012-10-17",
3    "Statement": [
4      {
5        "Effect": "Allow",
6        "Action": "ec2:*",
7        "Resource": "*",
8        "Condition": {
9          "StringEquals": {
10             "ec2:ResourceTag/Env": "development"
11          }
12        }
13      },
14      {
15        "Effect": "Allow",
16        "Action": "ec2:Describe*",
17        "Resource": "*"
18      },
19      {
20        "Effect": "Deny",
21        "Action": [
22          "ec2:DeleteTags",
23          "ec2:CreateTags"
24        ],
25        "Resource": "*"
26      }
27    ]
28  }
29
```



Account Alias

An account alias is simply a nickname for AWS account. Instead of a long account ID, I can now reference my account alias instead.

Creating an account alias took me 30 seconds - it's simple configuration in the IAM dashboard. Now, my new AWS console sign-in URL uses the alias instead of my account ID.

The screenshot shows a modal dialog titled "Create alias for AWS account 775412354186". It contains a text input field for the "Preferred alias" with the value "nextwork-alias-amirhamzah". Below the input is a note: "Must be not more than 63 characters. Valid characters are a-z, 0-9, and - (hyphen)." The "New sign-in URL" is displayed as "https://nextwork-alias-amirhamzah.signin.aws.amazon.com/console". A light blue information box states: "IAM users will still be able to use the default URL containing the AWS account ID." At the bottom are "Cancel" and "Create alias" buttons.

Create alias for AWS account 775412354186 ✕

Preferred alias

nextwork-alias-amirhamzah

Must be not more than 63 characters. Valid characters are a-z, 0-9, and - (hyphen).

New sign-in URL

https://nextwork-alias-amirhamzah.signin.aws.amazon.com/console

ⓘ IAM users will still be able to use the default URL containing the AWS account ID.

Cancel Create alias



IAM Users and User Groups

Users

IAM users are people or entities that have access/can login to our AWS account.

User Groups

IAM user groups are like folders that collect IAM users so that you can apply permission settings at the group level.

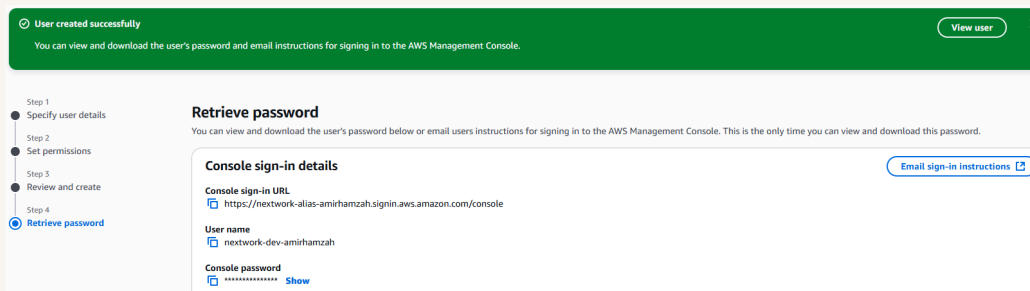
I attached the policy I created to this user group, which means any user created inside this group will automatically get the permissions attached to my NextWorkDevEnvironmentPolicy IAM Policy.



Logging in as an IAM User

The first way is to email sign-in instructions to the user, while the second way is to download a .csv file with the sign in details inside.

Once I logged in as my IAM user, I noticed that our user is already denied access to panels on the main AWS console dashboard. This was because we only setup permissions to our development EC2 instance.



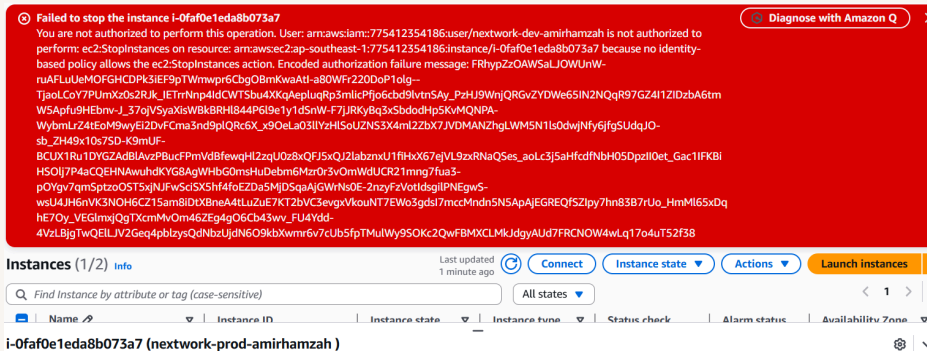


Testing IAM Policies

I tested my JSON IAM policy by attempting to stop both the development and production instance.

Stopping the production instance

When I tried to stop the production instance, I met with an error! This was because my production instance is tagged with the production label, which is outside of the scope of my permission policy.





Testing IAM Policies

Stopping the development instance

Next, when I tried to stop the development instance, I successfully saw the instance state change to Stopping and then Stopped. This was because my permission policy allows the intern to stop the instance.

Instances (1/2) Info		Last updated 1 minute ago		Connect
Find Instance by attribute or tag (case-sensitive) All states ▼				
	Name	Instance ID	Instance state	Instance type
☒	nextwork-dev-amirhamzah	i-03785ce55cd8de74b	Stopped	t2.micro
☐	nextwork-prod-amirhamzah	i-0faf0e1eda8b073a7	Running	t2.micro



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